📊 Data Cleaning in SQL – Complete Easy Notes

🔷 What is Data Cleaning?

Data Cleaning is the process of:
✓ Correcting errors
✓ Handling missing values
✓ Removing duplicates
✓ Fixing inconsistencies

👉 It prepares raw data for analysis or machine learning.

🔷 Why is Data Cleaning Important?

✓ Improves data quality
✓ Reduces errors in reports
✓ Makes analysis reliable
✓ Required before ML models

Clean data = Accurate insights 📈

[redacted] Step-by-Step Data Cleaning Process in SQL

We clean the table: messy_indian_dataset

✅ Step 1: Create and Populate Messy Dataset

Create table and insert messy data (already done earlier).

✂️ Step 2: Trim and Normalize Text Fields

🔷 Problem:

- Extra spaces
- Mixed uppercase & lowercase

🔷 Solution:

Use TRIM() and LOWER()

```
UPDATE messy_indian_dataset

SET name = TRIM(LOWER(name));
```

Repeat for:

- gender

- city
- state

🔷 **Functions Used**

Function	Purpose
TRIM()	Removes extra spaces
LOWER()	Converts to lowercase

🗓️ **Step 3: Clean Email & Phone Number**

🔷 **Clean Email**

Remove invalid characters:

```
UPDATE messy_indian_dataset
SET email = TRIM(REGEXP_REPLACE(LOWER(email), '[^a-z0-9@.]+',''));
```

👉 Removes unwanted symbols.

🔷 **Clean Phone Number**

Keep only digits:

```
UPDATE messy_indian_dataset
SET phone_number = REGEXP_REPLACE(phone_number, '[^0-9]+','');
```

🔷 **Function Used**

Function	Purpose
REGEXP_REPLACE()	Replaces unwanted patterns

🕵️ **Step 4: Extract Username from Email**

Add new column:

```
ALTER TABLE messy_indian_dataset
ADD COLUMN username VARCHAR(30);
```

Extract username:

```
UPDATE messy_indian_dataset
```

SET username = SUBSTRING_INDEX(email, '@', 1);

👉 Extracts text before @

📍 **Step 5: Create Location Field**

Add column:

ALTER TABLE messy_indian_dataset

ADD COLUMN location VARCHAR(50);

Concatenate city & state:

UPDATE messy_indian_dataset

SET location = CONCAT(city, ', ', state);

🔷 **Function Used**

Function	Purpose
CONCAT()	Joins text values

💰 **Step 6: Calculate Amount Without GST**

Add column:

ALTER TABLE messy_indian_dataset

ADD COLUMN without_gst DECIMAL(10,2);

Calculate 82% of purchase amount:

UPDATE messy_indian_dataset

SET without_gst = purchase_amount * 0.82;

📅 **Step 7: Add Expiry Date**

Add column:

ALTER TABLE messy_indian_dataset

ADD COLUMN exp_date DATE;

Add 3 years to purchase_date:

UPDATE messy_indian_dataset

SET exp_date = DATE_ADD(purchase_date, INTERVAL 3 YEAR);

🔷 **Function Used**

Function	Purpose
DATE_ADD()	Adds time interval

🚫 **Step 8: Remove Invalid Phone Numbers**

Delete rows where:

- Phone number ≠ 10 digits
- Contains non-numeric characters

```
DELETE FROM messy_indian_dataset
WHERE LENGTH(phone_number) != 10
  OR phone_number REGEXP '[^0-9]';
```

🚻 **Step 9: Standardize Gender Field**

Convert values to:

- 'M'
- 'F'
- 'other'

```
UPDATE messy_indian_dataset
SET gender = CASE
        WHEN gender IN ('m', 'male') THEN 'M'
        WHEN gender IN ('f', 'female') THEN 'F'
        ELSE 'other'
       END;
```

🔷 **Function Used**

Function	Purpose
CASE	Conditional logic

🍪 **Step 10: Validate Age Field**

Remove unrealistic ages:

```
DELETE FROM messy_indian_dataset
WHERE age <= 0 OR age >= 100;
```

👉 Keeps realistic age values only.

🎯 Summary of Data Cleaning Actions

Type of Cleaning	What We Did
Text Cleaning	TRIM, LOWER
Pattern Cleaning	REGEXP_REPLACE
Feature Engineering	username, location
Calculation	without_gst
Date Manipulation	exp_date
Validation	Phone number, age
Standardization	Gender values

🚀 Final Result

After all steps, the dataset is:

✓ Clean
✓ Consistent
✓ Valid
✓ Ready for analysis

🧠 Important SQL Functions Used

- TRIM()
- LOWER()
- REGEXP_REPLACE()
- CONCAT()
- SUBSTRING_INDEX()
- DATE_ADD()
- LENGTH()

- CASE

💡 **Final Key Concept**

Data Cleaning is not just removing NULL values.

It includes:
✓ Standardization
✓ Validation
✓ Transformation
✓ Feature creation
✓ Removing incorrect data

Good Data Cleaning = Strong Data Foundation 📊